

## Chapter 01 -- HTML Fundamentals

# Practice Set

25 Hands-On HTML Coding Exercises

Section A -- Beginner (Q1-Q10)

Section B -- Intermediate  
(Q11-Q20)

Section C -- Challenge (Q21-Q25)

### Instructions

1. Each question requires writing actual HTML code in a .html file. 2. Save your answer using the filename given at the end of each question. 3. Open each file in a browser to verify your output visually. 4. Write clean, properly indented, valid HTML throughout. 5. Always include the full HTML5 boilerplate unless told otherwise. 6. Validate your work at <https://validator.w3.org> before marking done.

## Section A -- Beginner Questions 1 through 10

These questions cover core HTML fundamentals. Complete each one independently without referring to notes.

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### Question 1: HTML5 Boilerplate from Memory

Without any reference material, create a complete valid HTML5 page from memory. Your page MUST include: (1) the DOCTYPE declaration, (2) the html element with a lang attribute set to English, (3) a complete head section containing meta charset, meta viewport, a meaningful title tag, and a link to an external CSS file called style.css, (4) a body section with a main heading (h1), two paragraphs of placeholder text, and a horizontal rule between them. Do not copy from any source -- type it all from memory.

Save your answer as: q01.html

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### Question 2: All Six Heading Levels on One Page

Create an HTML page that uses all six heading levels (h1 through h6) on a single page. Each heading must display the text "Heading Level [number] - [example use case]" so the purpose of each level is clear. After all six headings, add a paragraph explaining when and why each heading level should be used. Remember the rule: only ONE h1 tag per page.

Save your answer as: q02.html

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### Question 3: Text Formatting Showcase Page

Create a paragraph-based article page on any topic of your choice. Your article MUST use ALL of the following text formatting tags meaningfully in natural context: strong, em, b, i, mark, del, ins, sup, and sub. Additionally include one pre element and one blockquote element. Each tag must be used because it is contextually appropriate -- not just to satisfy the requirement. The text must read naturally.

Save your answer as: q03.html

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### Question 4: Unordered Shopping List with Categories

Create a shopping list page using unordered lists. Requirements: an h1 "Weekly Shopping List", at least 3 category headings (use h2: Produce, Dairy, Pantry), each with its own unordered list containing at least 5 items. Use a third-level h3 for at least one sub-category within a section (e.g., "Organic" vs "Conventional" in Produce). Add a short introductory paragraph and a total item count at the bottom.

Save your answer as: q04.html

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### Question 5: Step-by-Step Instructions with Ordered List

Create an instructional page for any real-world process (e.g., how to make coffee, how to restart a router, how to tie shoelaces). Include: an h1 with the process name, an introduction paragraph, an ordered list of at least 10 detailed steps with the default number style, a "What You Will Need" unordered list before the steps, and a conclusion paragraph after the steps. Also add one nested ordered list within a step that itself

has sub-steps.

Save your answer as: q05.html

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### Question 6: Image Page with Proper Alt Text

Create an HTML page with a gallery of exactly 6 images. Use free placeholder images from: <https://picsum.photos/400/300?random=1> (change the number for each image). For EACH image: write a descriptive alt attribute (not just "image" or a filename), include explicit width="400" and height="300" attributes, add loading="lazy" to images 2 through 6, wrap each image in a figure element with a figcaption below it. Add a page title (h1) and a short intro paragraph.

Save your answer as: q06.html

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### Question 7: External Links with Full Security Attributes

Create a "Useful Developer Resources" page with links to exactly 6 real websites: MDN Web Docs, W3Schools, CSS-Tricks, GitHub, Stack Overflow, and Can I Use. Every link MUST: open in a new tab (target="\_blank"), include rel="noopener noreferrer" for security, have meaningful descriptive link text (not "click here"), and be wrapped in a list item inside an unordered list. Organize the links under an h1 heading and include a brief description of each resource as a sentence after the link.

Save your answer as: q07.html

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### Question 8: Contact Page with Mailto, Tel, and Address

Create a contact page for a fictional business called "Pixel Perfect Studio". The page must include: the business name as h1, a tagline as a paragraph, an address section using the HTML address element with street, city, state, zip, a clickable email link using mailto: (use info@pixelperfect.example), a clickable phone link using tel: with a full number including country code, links to 3 fictional social media profiles (use # as href), and business hours displayed as a definition list (dl, dt, dd).

Save your answer as: q08.html

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### Question 9: Nested List -- Course Curriculum

Create a web development course curriculum page using nested lists. Structure: use an ordered list for 4 modules (Module 1 through 4). Inside each module list item, include the module title and then a nested ordered list of at least 4 lesson topics. Inside at least 2 lessons, add a third level of nesting -- an unordered list of sub-topics or learning objectives for that lesson. Add a page title (h1), an intro paragraph, and a total lesson count at the bottom.

Save your answer as: q09.html

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### Question 10: Single-Page Site with Anchor Link Navigation

Create a long single-page website for a fictional organization, portfolio, or product of your choice. Requirements: a navigation bar at the TOP of the page (use nav element) with at least 5 anchor links pointing to sections below using #id; at least 5 content sections each with a matching id attribute, a h2 heading, at least 2 paragraphs of content, and one list; a "Back to Top" link at the bottom of EVERY section; a footer with copyright information. The navigation must use an unordered list inside the nav element.

Save your answer as: q10.html

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## Section B -- Intermediate Questions 11 through 20

These questions require combining multiple HTML concepts and applying them to realistic, practical scenarios.

### Question 11: Web Development Glossary with Definition Lists

Build a full glossary page for web development terminology using HTML definition lists. Requirements: an h1 "Web Development Glossary", a linked table of contents at the top with anchor links to each category; at least 4 categories (e.g., HTML Terms, CSS Terms, JavaScript Terms, General Terms) each as an h2 with a unique id; at least 5 terms per category using dl, dt, and dd; each definition must be at least 2 complete sentences explaining the term clearly; a "Back to Top" link after each category; use abbr elements for any abbreviations you reference.

Save your answer as: `q11.html`

### Question 12: Semantic Text Formatting Article

Write a short article (fiction or non-fiction, minimum 5 paragraphs) that uses ALL of the following tags meaningfully and in appropriate context: strong, em, b, i, mark, del, ins, sup, sub, abbr, cite, blockquote, q, code, pre, kbd, samp, and var. Each tag must be used because it is semantically correct for that content -- not forced in randomly. Add an HTML comment directly above each tag explaining in one sentence WHY you used that specific tag at that point.

Save your answer as: `q12.html`

### Question 13: Structured Page Layout with div and span

Build a webpage with a clear layout structure using div and span elements with descriptive class names (no CSS required). Required div sections (each with a class): site-header, site-nav, page-hero, main-content, content-article (at least 2), content-sidebar, site-footer. Add an HTML comment above EACH div explaining its purpose and content. Inside at least 4 paragraphs, use span elements with class attributes to highlight or group specific words or phrases. All content must be meaningful (not lorem ipsum).

Save your answer as: `q13.html`

### Question 14: Semantic HTML Image Gallery

Build a complete image gallery page using HTML5 semantic elements. Requirements: an h1 gallery title, a short intro paragraph, exactly 9 images arranged conceptually in 3 thematic groups (use h2 for each group title, e.g., "Nature", "Architecture", "Abstract"), each image must use the figure + figcaption structure, alt text must be specific and descriptive for each image, include width and height for each image, use loading="lazy" on all images except the first 3, use picsum.photos for placeholder images with different seeds.

Save your answer as: `q14.html`

### Question 15: Full Restaurant Menu with Definition Lists

Build a complete restaurant menu for a fictional restaurant of your own invention. Requirements: restaurant name as h1, a tagline and description paragraph, at least 5 menu sections (Starters, Salads, Mains, Desserts, Drinks) each as h2; each section uses a definition list: dish name as dt, description + price in the dd; at least 4 dishes per section; use del to show an original price and ins for a current price on at least 3 dishes; an allergen notice section using an unordered list; add a note about reservations using an address element.

Save your answer as: `q15.html`

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### Question 16: Personal Profile Page

Build a complete personal profile page (real or entirely fictional). Required elements: a profile photo (use `picsum.photos`) with proper alt text wrapped in a figure with `figcaption`; full name as h1; a professional title or tagline as h2; a bio of at least 5 sentences; an "Education" section with a definition list (school, degree, year); a "Skills" section with an unordered list of at least 8 skills; a "Hobbies" section with an unordered list; a "Connect" section with links to at least 4 social/professional profiles (email, LinkedIn, GitHub, Twitter/X -- use # or mailto: as needed); a footer with a copyright line.

Save your answer as: `q16.html`

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### Question 17: Resource Hub with Internal and External Links

Build a "Full-Stack Developer Resources" hub page combining both link types. Requirements: an h1 and intro paragraph; a table of contents at the top with anchor links to at least 6 sections below; each section (h2 with unique id) covers a topic: HTML, CSS, JavaScript, React, Backend, and Tools; each section must contain at least 4 external links to real resources, each opening in a new tab with `rel="noopener noreferrer"`, each with a short description after the link; a "Back to Table of Contents" link at the bottom of each section.

Save your answer as: `q17.html`

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### Question 18: Book Library Catalog Page

Create a library catalog page featuring exactly 10 books. Group the books into at least 3 genre categories (e.g., Fiction, Non-Fiction, Technical) using h2 headings. For each book include: book title as h3, a definition list with Author, Year, Genre, Publisher, Pages, and ISBN as terms; a brief description paragraph (2-3 sentences); use del for an old price and ins for the current price on at least 5 books; use sup for at least one edition marker (e.g., 2nd edition); add an "Add to Reading List" anchor link for each book (use # as href).

Save your answer as: `q18.html`

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### Question 19: All Void Elements Documented Showcase

Create a reference page that uses every HTML void element at least once, each with an HTML comment above it explaining what it does. Required void elements: area (inside a map with usemap), base (in head), br (in a paragraph), col (inside colgroup inside table), embed, hr, img (at least 3 with proper alt), input with at least 6 different type values: text, email, password, checkbox, radio, submit, link (in head, at least 2 different rel values), meta (at least 4 different uses: charset, viewport, description, author), source (inside a video or audio element), track (inside a video element), and wbr (inside a long word). Organize everything into labeled sections with h2 headings.

Save your answer as: `q19.html`

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**Question 20: News Homepage with Semantic HTML and BEM Class Names**

Build a news website homepage using proper semantic HTML and BEM-style class names. Required semantic elements: header, nav, main, aside, footer, at least 3 article elements, at least 2 section elements, figure + figcaption for at least 2 images. Class naming must follow BEM convention:

block\_\_element--modifier (e.g., class="news-card\_\_title", class="nav\_\_link--active"). Content must be realistic news content (real or fictional). Add HTML comments explaining the purpose of each major section. The page should have a clear visual hierarchy using only HTML headings.

Save your answer as: q20.html

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## Section C -- Challenge Questions 21 through 25

These questions require significant planning, research, and effort. They test your ability to combine all Chapter 01 concepts into real-world projects.

### Question 21: Multi-Page Website with Shared Navigation

Build a complete 3-page mini website using 3 separate HTML files that link to each other. File 1 - q21-index.html (Homepage): A hero section with h1 and tagline paragraph; a "Features" section (h2) with an unordered list of at least 6 features; a "Latest News" section (h2) with 3 article elements each containing a headline (h3), date, author, and 2-paragraph body. File 2 - q21-about.html (About Page): A company story section with h1 and 3 paragraphs; a "Our Team" section (h2) with at least 3 team member profiles -- each with a photo (figure + figcaption), name (h3), role (h4), and bio paragraph. File 3 - q21-contact.html (Contact Page): A contact h1; a full address block (address element); a phone link and email link; 4 social media links in an unordered list; a "Business Hours" section using a definition list. ALL 3 PAGES MUST HAVE: an identical navigation bar (nav with ul and li) with working links to all 3 pages; a site header with the organization name; a footer with copyright and navigation links. The current page link in the nav should have class="active".

Save your answer as: q21-index.html + q21-about.html + q21-contact.html

### Question 22: Newspaper Front Page with Pure HTML

Recreate the front page structure of a newspaper using only HTML structural elements (no CSS styling beyond default browser rendering). Required elements: newspaper name as h1 with publication date and edition as a paragraph; a horizontal rule after the masthead; a "Top Stories" section with at least 3 complete news articles, each with: a headline (h2), dateline and byline (using cite and time), at least 3 paragraphs of article text, an image with caption (figure + figcaption), and a "Continued on page X" note; a sidebar section (aside) with: weather (use a definition list), stock market summary (definition list), and "Letters to the Editor" with at least 2 blockquotes with cite attributes; a footer with editorial staff credits and contact information (address element).

Save your answer as: q22.html

### Question 23: Complete Recipe Page

Build a comprehensive recipe page covering every standard recipe component. Required sections: - Recipe title as h1 - Introduction: a paragraph describing the dish and its origin (use cite for sources) - Recipe Metadata (h2): definition list with prep time, cook time, total time, servings, difficulty, and cuisine type - Ingredients (h2): at minimum 2 ingredient groups (e.g., "For the Base:", "For the Sauce:") each as an h3 with an unordered list of at least 6 items; use sub for any measurements with subscripts; use mark to highlight key ingredients - Instructions (h2): ordered list with at least 12 detailed steps; use strong for critical warnings (e.g., "Do not overcook"); use em for temperature indications - Chef Tips (h2): unordered list of at least 5 tips - Nutritional Info (h2): definition list with at least 8 nutritional facts - Reader Reviews (h2): at least 3 blockquotes with author (cite) and star rating - Related Recipes (h2): unordered list of links to 4 other recipes (use # as href)

Save your answer as: q23.html



### Question 24: Every Chapter 01 Tag Used and Documented

Create a single comprehensive reference page that uses EVERY HTML tag covered in Chapter 01 at least once. Above every single tag occurrence, add an HTML comment saying what it does and why you are using it at that point. Required tags (minimum, all must appear): !DOCTYPE, html (with lang), head, meta (charset, viewport, description, author), title, link (stylesheet and icon), body, header, main, footer, nav, section (at least 3), article (at least 2), aside, div (at least 3), span (at least 3), h1, h2, h3, h4, h5, h6, p (at least 6), br, hr, strong, em, b, i, mark, del, ins, sup, sub, pre, code, a (with href, mailto, tel, target, rel, and download versions), img (with src, alt, width, height, and loading attributes), ul (at least 2), ol (at least 2), li, dl, dt, dd, blockquote (with cite), cite, abbr, address, figure, figcaption, table, thead, tbody, tfoot, tr, th, td, caption. Organize the page into clear topic sections using h2 headings. The page must be valid HTML5 with no errors at validator.w3.org.

Save your answer as: `q24.html`

### Question 25: Working Linked Table of Contents -- Long-Form Document

Build a comprehensive long-form HTML document on any substantial topic of your choice (minimum 2,500 words of real, meaningful content -- not lorem ipsum). Required structure: - A fixed/sticky header with the document title (h1) - A "Table of Contents" section at the top with: an h2 "Table of Contents", a numbered ordered list of ALL main sections AND their sub-sections as nested ordered lists; every list item must be an anchor link with href pointing to the matching section id - At least 6 main sections (h2 with unique id) - Each main section must contain: at least 3 sub-sections (h3, each with unique id), at least 3 paragraphs of content, one unordered or ordered list, one image with proper alt text (use picsum.photos), one code or blockquote element where relevant, a horizontal rule before the navigation links, a "Back to Top" link AND a "Back to Table of Contents" link - Each sub-section must have: an h3 with unique id, at least 2 paragraphs - A concluding footer section with author, date (use time element), and references (ordered list of citations using cite elements) - Total minimum: 6 sections x 3 sub-sections = 18 headings linked from the TOC

Save your answer as: `q25.html`